



VIT[®]
B H O P A L
www.vitbhopal.ac.in

VIT BHOPAL UNIVERSITY

School of Computing Science and Engineering

Bhopal-Indore Highway, Kothrikalan, Sehore

Madhya Pradesh - 466114

CSE3016 – AWS SOLUTION ARCHITECT

REG.NO : 23BAI10917

NAME : Vijay Rajesh R

BRANCH : AIML

SEMESTER: INTERIM SEMESTER 2025-26

INDEX

Ex.No	DATE	EXPERIMENT NAME	PAGE NO.
1.	10-11-25	Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware.	03
2.	11-11-25	Deploy a static website in S3 and manage the website using S3 life cycle management.	10
3.	12-11-25	Creating DNS records, configuring routing policies, implementing health checks, and monitoring DNS health using Route 53.	14
4.	13-11-25	Create and configure Amazon Elastic File System (EFS) to create scalable and shared file storage for AWS resources and EFS file system, mount it to EC2 instances, and configure access permissions.	
5.	13-11-25	Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience.	
6.	14-11-25	Create AWS Certificate Manager (ACM) for managing SSL/TLS certificates for secure communication in AWS.	
7.	14-11-25	Create and manage customer master keys (CMKs), encrypt data using KMS, and integrate KMS with AWS services for encryption.	
8.	14-11-25	Set up CloudWatch monitoring for AWS resources, create custom metrics and alarms, analyze logs, and troubleshoot performance issues using CloudWatch.	

EXP.NO: 01	Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware
DATE: 10-11-25	

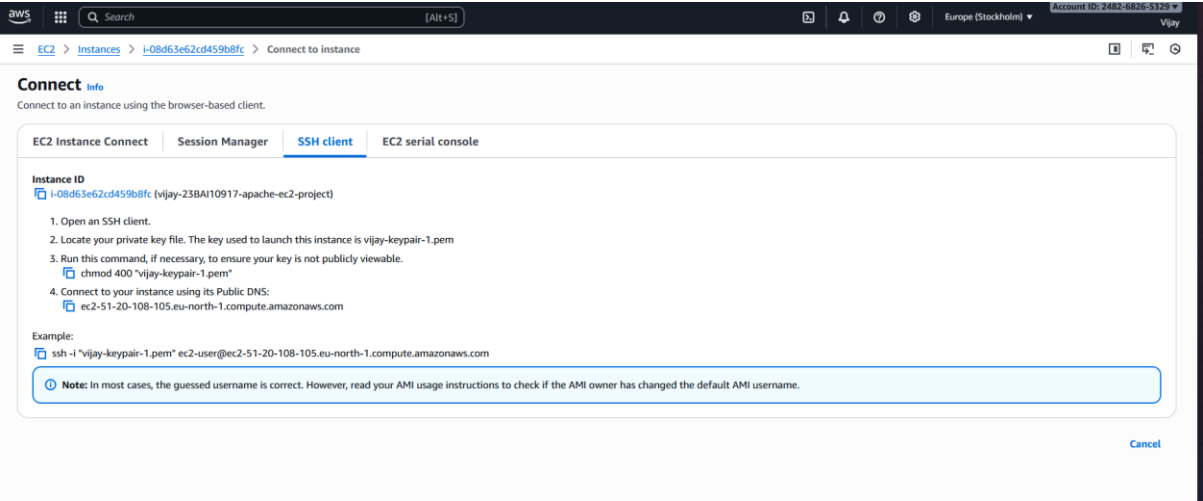
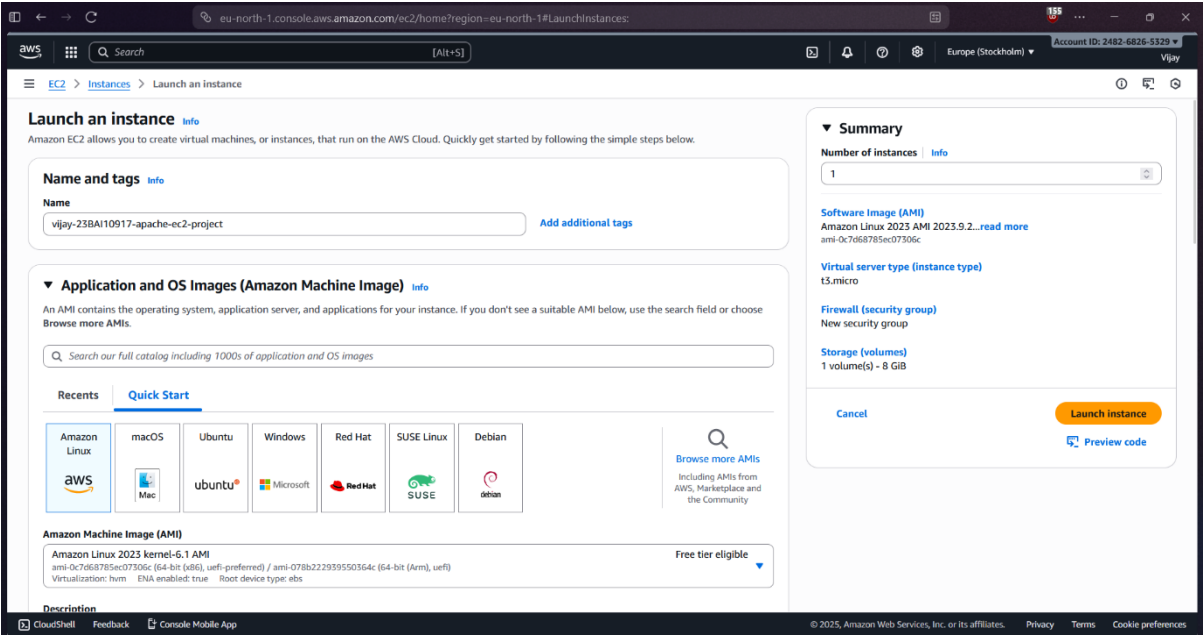
AIM

To Launch an EC2 instance using the AWS Management Console and to successfully deploy a website using Apache middle ware

PROCEDURE

1. Launch EC2 Instance: Use the AWS Console to launch a t2.micro instance with Amazon Linux 2023 and an appropriate Key Pair.
2. Configure Security Group: Ensure the instance's Security Group allows inbound traffic on Port 22 (SSH) and Port 80 (HTTP).
3. Connect via SSH: Use your downloaded .pem key file to SSH into the instance using the public IP address.
4. Install Apache: Run `sudo dnf update -y` then `sudo dnf install httpd -y` on the instance.
5. Start and Enable Service: Run `sudo systemctl start httpd` and `sudo systemctl enable httpd`.
6. Deploy Website: Navigate to `/var/www/html/` and create or upload your website files (e.g., `sudo nano index.html`).
7. Test: Open the instance's Public IPv4 Address in a web browser to view the deployed website.

IMPLEMENTATION SCREENSHOT




```

[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64
Amazon Linux 2023 Kernel Livepatch repo 273 kB/s | 29 kB    00:00
Dependencies resolved.
=====
Package                Arch    Version                               Repository    Size
=====
Installing:
httpd                  x86_64  2.4.65-1.amzn2023.0.2                amazonlinux   47 k
Installing dependencies:
apr                    x86_64  1.7.5-1.amzn2023.0.4                amazonlinux   129 k
apr-util              x86_64  1.6.3-1.amzn2023.0.2                amazonlinux   97 k
apr-util-ldb          x86_64  1.6.3-1.amzn2023.0.2                amazonlinux   13 k
generic-logos-httpd  noarch  18.0.0-12.amzn2023.0.3              amazonlinux   19 k
httpd-core            x86_64  2.4.65-1.amzn2023.0.2                amazonlinux  1.4 M
httpd-filesystem      noarch  2.4.65-1.amzn2023.0.2                amazonlinux   13 k
httpd-tools           x86_64  2.4.65-1.amzn2023.0.2                amazonlinux   81 k
libbrotli             x86_64  1.0.9-4.amzn2023.0.2                amazonlinux  315 k
mailcap               noarch  2.1.49-3.amzn2023.0.3                amazonlinux   33 k
Installing weak dependencies:
apr-util-openssl      x86_64  1.6.3-1.amzn2023.0.2                amazonlinux   15 k
mod_http2             x86_64  2.0.27-1.amzn2023.0.3                amazonlinux  166 k
mod_lua               x86_64  2.4.65-1.amzn2023.0.2                amazonlinux   60 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-ldb-1.6.3-1.amzn2023. 369 kB/s | 13 kB    00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64 3.0 MB/s | 129 kB    00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x 2.1 MB/s | 97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn20 737 kB/s | 15 kB     00:00
(5/13): httpd-2.4.65-1.amzn2023.0.2.x86 2.5 MB/s | 47 kB     00:00
(6/13): generic-logos-httpd-18.0.0-12.a 786 kB/s | 19 kB     00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2 578 kB/s | 13 kB     00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0 2.9 MB/s | 81 kB     00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0. 28 MB/s | 1.4 MB    00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2 9.9 MB/s | 315 kB    00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3. 1.3 MB/s | 33 kB     00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0. 6.4 MB/s | 166 kB    00:00
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s | 60 kB     00:00
-----
Total                                14 MB/s | 2.4 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Installing     : apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_ 3/13
  Installing     : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Installing     : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13

```

```

(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s | 60 kB      00:00
-----
Total                               14 MB/s | 2.4 MB      00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Installing     : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
  Installing     : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Installing     : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
  Installing     : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 6/13
  Installing     : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
  Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Installing     : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Installing     : httpd-core-2.4.65-1.amzn2023.0.2.x86_64 9/13
  Installing     : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
  Installing     : mod_lua-2.4.65-1.amzn2023.0.2.x86_64 11/13
  Installing     : generic-logos-httpd-18.0.0-12.amzn2023.0.3 12/13
  Installing     : httpd-2.4.65-1.amzn2023.0.2.x86_64 13/13
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64 13/13
  Verifying      : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
  Verifying      : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
  Verifying      : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13
  Verifying      : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
  Verifying      : generic-logos-httpd-18.0.0-12.amzn2023.0.3 5/13
  Verifying      : httpd-2.4.65-1.amzn2023.0.2.x86_64 6/13
  Verifying      : httpd-core-2.4.65-1.amzn2023.0.2.x86_64 7/13
  Verifying      : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa 8/13
  Verifying      : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 9/13
  Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
  Verifying      : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
  Verifying      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
  Verifying      : mod_lua-2.4.65-1.amzn2023.0.2.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.65-1.amzn2023.0.2.x86_64
httpd-core-2.4.65-1.amzn2023.0.2.x86_64
httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
httpd-tools-2.4.65-1.amzn2023.0.2.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-31-177 ec2-user]# |

```

```
Complete!
[root@ip-172-31-31-177 ec2-user]# systemctl start httpd.service
[root@ip-172-31-31-177 ec2-user]# systemctl enable start httpd.service
Failed to enable unit: Unit file start.service does not exist.
[root@ip-172-31-31-177 ec2-user]# systemctl enable httpd.service
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-31-177 ec2-user]#
```

The screenshot displays the AWS Management Console interface for an EC2 instance. The top navigation bar shows the AWS logo, a search bar, and the user's account information (Vijay, Account ID: 2482-6826-5329, Europe (Stockholm)). The left sidebar contains navigation links for EC2, Instances, and various EC2 services like Dashboard, AWS Global View, Events, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, and Lifecycle Manager.

The main content area is titled "Instances (1/1) Info" and shows a table with one instance: "vijay-23BAI10917-apache-ec2-project". The instance is in the "Running" state, with a status check of "Initializing". The instance type is "t3.micro" and it is located in the "eu-north-1a" availability zone. The public IPv4 address is "ec2-51-20-108-105.eu-north-1.compute.amazonaws.com".

The instance details are expanded, showing the "Instance summary" tab. The instance ID is "i-08d63e62cd459b8fc". The instance state is "Running". The public IPv4 address is "51.20.108.105". The private IPv4 address is "172.31.31.177". The public DNS is "ec2-51-20-108-105.eu-north-1.compute.amazonaws.com". The private IP DNS name (IPv4 only) is "ip-172-31-31-177.eu-north-1.compute.internal".

Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the Apache HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting `www.example.com`, you should send e-mail to `"webmaster@example.com"`.

If you are the website administrator:

You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

You are free to use the image below on web sites powered by the Apache HTTP Server:



RESULT

Thus, EC2 instance has been launched using the AWS Management Console and a website is deployed using the Apache Middle ware

EXP.NO: 02	Deploy a static website in S3 and manage the website using S3 life cycle management
DATE: 11-11-25	

AIM

To deploy a static website in S3 and manage the website using S3 Life Cycle Management, in the AWS Management Console

PROCEDURE

Step 1: Create and Configure the S3 Bucket

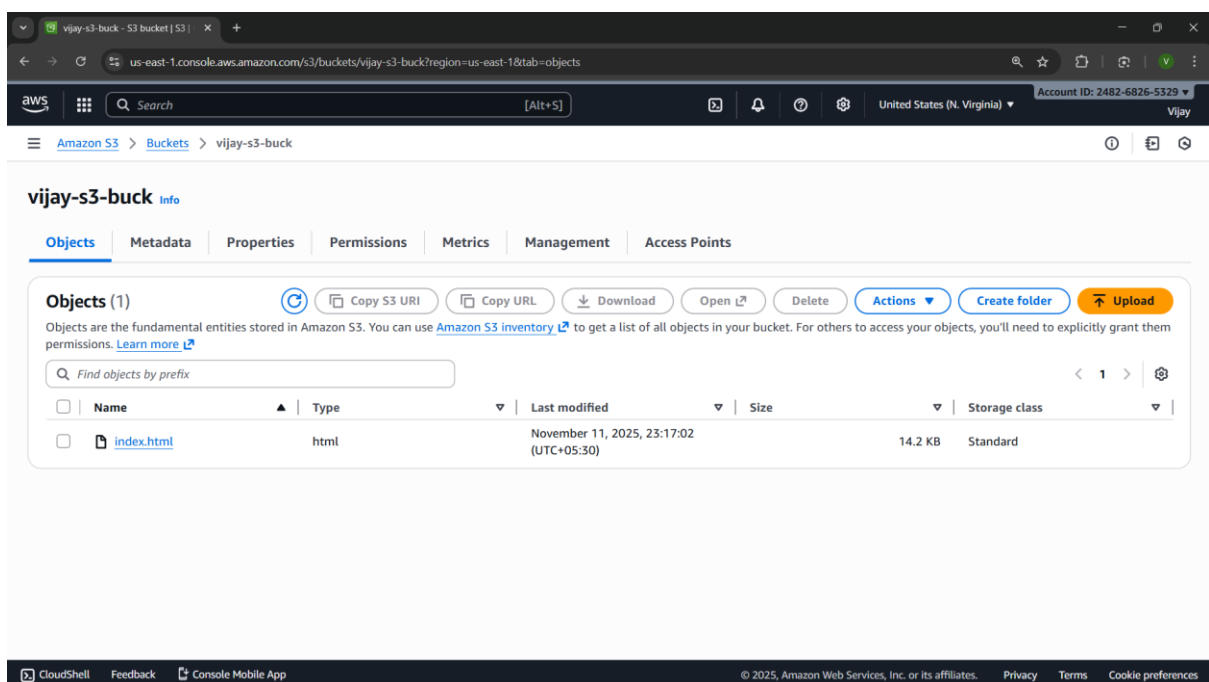
- Create Bucket: In the AWS S3 Console, click "Create bucket". Give it a globally unique name (e.g., my-static-site).
- Public Access: Uncheck "Block all public access" during creation (you'll set permissions next).
- Enable Website Hosting: Go to the bucket's Properties tab, find Static website hosting, click "Edit", select "Enable", and specify your Index document (e.g., index.html) and Error document (e.g., error.html). Save changes.

Step 2: Upload Files and Set Permissions

- Upload Content: Go to the Objects tab and Upload all your website files (HTML, CSS, images).
- Apply Bucket Policy: Go to the Permissions tab and click "Edit" under Bucket Policy. Paste a policy granting public s3:GetObject access to all resources in your bucket, replacing YOUR_BUCKET_NAME. Save changes.
- Test Access: Paste the Bucket website endpoint (found under Properties) into a browser to confirm the site is live.
- Step 3: Implement Lifecycle Management
- Create Rule: Go to the Management tab, find Lifecycle rules, and click "Create lifecycle rule".

- Define Scope: Give the rule a name (e.g., Archive-and-Expire) and choose "Apply to all objects" or use a prefix (e.g., backups/) to target specific folders.
- Configure Transitions (Cost Optimization): Check "Transition current versions of objects...".
- Click "Add transition": Move objects to Standard-IA after 30 days.
- Click "Add transition": Move objects to Glacier Instant Retrieval after 90 days.
- Configure Expiration (Cleanup): Check "Expire current versions of objects" and set the number of days after creation (e.g., 365 days) to permanently delete the objects.
- Finalize: Click "Create rule". The lifecycle rule will now automatically manage the storage class and eventual deletion of your objects based on their age.

IMPLEMENTATION SCREENSHOT



Create Lifecycle rule - S3 bucket

us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule

Create lifecycle rule Info

Lifecycle rule configuration

Lifecycle rule name

vijay-lifecycle-rule-1

Up to 255 characters

Choose a rule scope

Limit the scope of this rule using one or more filters

Apply to all objects in the bucket

Apply to all objects in the bucket

If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

☒ Transition current versions of objects between storage classes

This action will move current versions.

☐ Transition noncurrent versions of objects between storage classes

CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Create Lifecycle rule - S3 bucket

us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1

Search [Alt+S]

United States (N. Virginia) Account ID: 2482-6826-5329 Vijay

Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule

Day 0

• Objects uploaded

↓

Day 30

• Objects move to Standard-IA

↓

Day 60

• Objects move to Intelligent-Tiering

↓

Day 90

• Objects move to One Zone-IA

↓

Day 240

• Objects move to Glacier Flexible Retrieval (formerly Glacier)

↓

Day 365

• Objects move to Glacier Deep Archive

Day 0

No actions defined.

Cancel

Create rule

CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Standard-IA	30	Remove
Intelligent-Tiering	60	Remove
One Zone-IA	90	Remove
Glacier Flexible Retrieval (formerly Glacier)	240	Remove
Glacier Deep Archive	365	Remove
Add transition		

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

	Lifecycle rule name	Status	Scope	Current version actions	Noncurrent versions ac...	Expired object delete ...
	vijay-lifecycle-rule-1	Enabled	Entire bucket	Transition to Standard-IA, then I	-	-

RESULT

Thus, a static website is deployed in S3 and it is managed using S3 life cycle management.

EXP.NO: 03	Creating DNS records, configuring routing policies, implementing health checks, and monitoring DNS health using Route 53
DATE: 12-11-25	

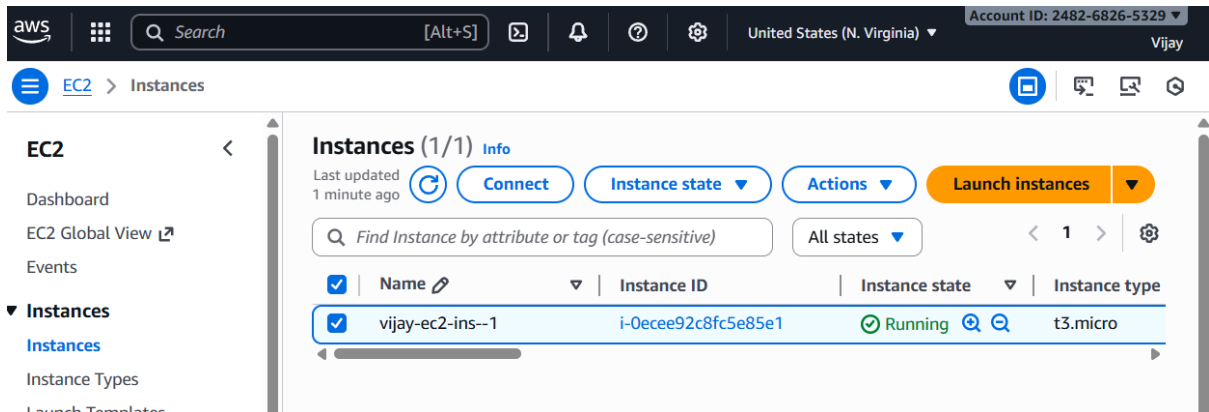
AIM

To establish a low-latency, and highly available Domain Name System (DNS) architecture for a web application using AWS Route 53, including the configuration of various routing policies, implementation of active health checks for continuous resource monitoring, and integration with CloudWatch for automated DNS health monitoring and alerting capabilities.

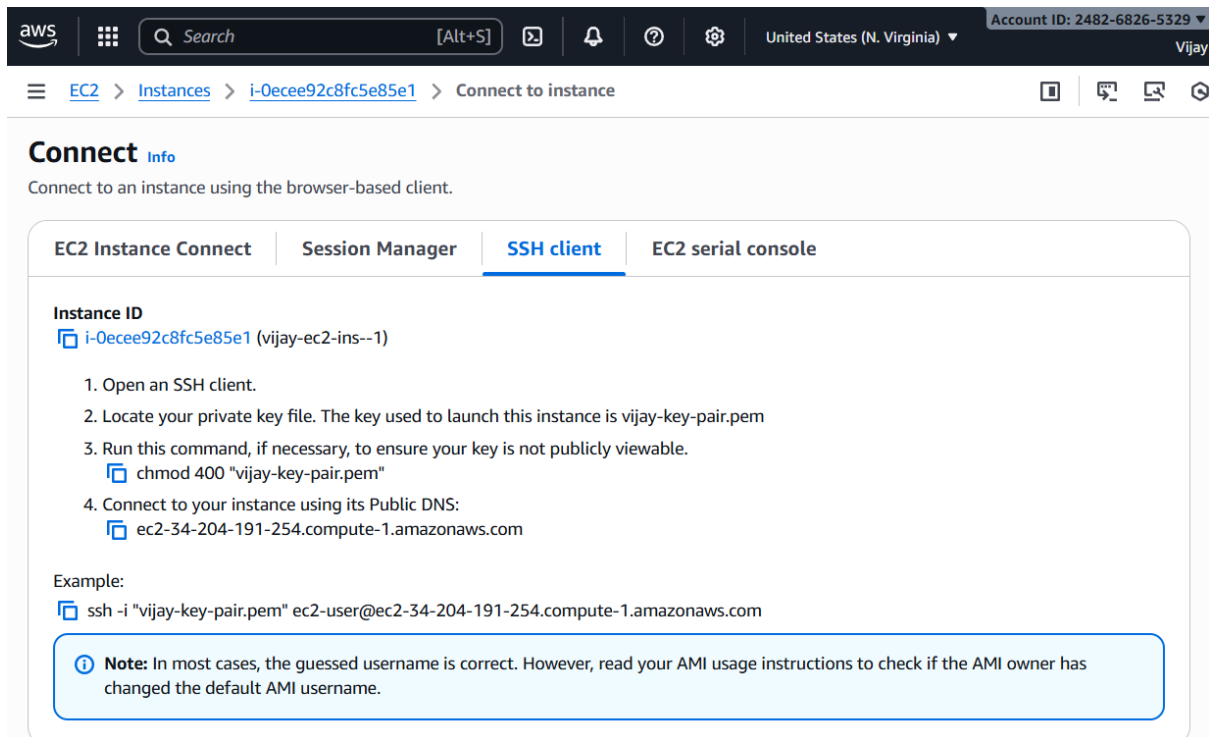
PROCEDURE

1. Create Hosted Zone: In Route 53, create a Public Hosted Zone for your domain (example.com).
2. Create Resource Records: Create an A record (or CNAME/ALIAS) pointing to your resource (e.g., EC2/Load Balancer IP). Set the Routing Policy (e.g., Simple or Failover).
3. Implement Health Check: Create a Health Check to monitor your endpoint (IP address, domain, port, and protocol, e.g., HTTP on port 80).
4. Configure Routing: For Failover or Weighted routing, edit the resource record and associate it with the new Health Check.
5. Set Monitoring/Alerts: Select the Health Check and create a CloudWatch Alarm (e.g., using the HealthCheckStatus metric) to send an SNS notification if the status becomes Unhealthy.

IMPLEMENTATION SCREENSHOT



The screenshot shows the AWS Management Console interface for the EC2 Instances page. The top navigation bar includes the AWS logo, a search bar, and the account ID '2482-6826-5329'. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'EC2 Global View', 'Events', 'Instances', 'Instance Types', and 'Launch Templates'. The main content area displays the 'Instances (1/1)' page, which includes a 'Connect' button, an 'Instance state' dropdown, and an 'Actions' dropdown. A table lists the instance 'vijay-ec2-ins--1' with ID 'i-0ecee92c8fc5e85e1', state 'Running', and type 't3.micro'.



The screenshot shows the 'Connect to instance' page in the AWS Management Console. The page title is 'Connect' and it includes an 'Info' link. The instructions state: 'Connect to an instance using the browser-based client.' The page is divided into four tabs: 'EC2 Instance Connect', 'Session Manager', 'SSH client', and 'EC2 serial console'. The 'SSH client' tab is selected. It displays the instance ID 'i-0ecee92c8fc5e85e1 (vijay-ec2-ins--1)' and provides a list of steps to connect via SSH. The steps are: 1. Open an SSH client. 2. Locate your private key file. The key used to launch this instance is vijay-key-pair.pem. 3. Run this command, if necessary, to ensure your key is not publicly viewable. 4. Connect to your instance using its Public DNS: ec2-34-204-191-254.compute-1.amazonaws.com. An example command is provided: ssh -i "vijay-key-pair.pem" ec2-user@ec2-34-204-191-254.compute-1.amazonaws.com. A note at the bottom states: 'Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.'


```
[root@ip-172-31-72-108 ec2-user]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository                               278 kB/s | 29 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
httpd                                   x86_64             2.4.65-1.amzn2023.0.2                 amazonlinux          47 k
Installing dependencies:
apr                                     x86_64             1.7.5-1.amzn2023.0.4                 amazonlinux          129 k
apr-util                               x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux          97 k
apr-util-ldap                          x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux          13 k
generic-logos-httpd                   noarch             18.0.0-12.amzn2023.0.3               amazonlinux          19 k
httpd-core                             x86_64             2.4.65-1.amzn2023.0.2                 amazonlinux          1.4 M
httpd-filesystem                       noarch             2.4.65-1.amzn2023.0.2                 amazonlinux          13 k
httpd-tools                            x86_64             2.4.65-1.amzn2023.0.2                 amazonlinux          81 k
libbrotli                              x86_64             1.0.9-4.amzn2023.0.2                 amazonlinux          315 k
mailcap                                noarch             2.1.49-3.amzn2023.0.3                 amazonlinux          33 k
Installing weak dependencies:
apr-util-openssl                       x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux          15 k
mod_http2                              x86_64             2.0.27-1.amzn2023.0.3                 amazonlinux          166 k
mod_lua                                x86_64             2.4.65-1.amzn2023.0.2                 amazonlinux          60 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                2.7 MB/s | 97 kB    00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm           346 kB/s | 13 kB    00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                     2.9 MB/s | 129 kB    00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm        734 kB/s | 15 kB    00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm   822 kB/s | 19 kB    00:00
(6/13): httpd-2.4.65-1.amzn2023.0.2.x86_64.rpm                 2.1 MB/s | 47 kB    00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch.rpm       604 kB/s | 13 kB    00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0.2.x86_64.rpm           3.5 MB/s | 81 kB    00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0.2.x86_64.rpm            33 MB/s | 1.4 MB    00:00
(10/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm               1.5 MB/s | 33 kB    00:00
(11/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm             8.5 MB/s | 315 kB   00:00
```

```
Installing      : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64      6/13
Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64       7/13
Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch 8/13
Installing      : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch 8/13
Installing      : httpd-core-2.4.65-1.amzn2023.0.2.x86_64      9/13
Installing      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64      10/13
Installing      : mod_lua-2.4.65-1.amzn2023.0.2.x86_64        11/13
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing      : httpd-2.4.65-1.amzn2023.0.2.x86_64          13/13
Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64          13/13
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64             1/13
Verifying       : apr-util-1.6.3-1.amzn2023.0.2.x86_64        2/13
Verifying       : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64   3/13
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying       : httpd-2.4.65-1.amzn2023.0.2.x86_64          6/13
Verifying       : httpd-core-2.4.65-1.amzn2023.0.2.x86_64     7/13
Verifying       : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch 8/13
Verifying       : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64     9/13
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64      10/13
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch         11/13
Verifying       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64     12/13
Verifying       : mod_lua-2.4.65-1.amzn2023.0.2.x86_64        13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64                                apr-util-1.6.3-1.amzn2023.0.2.x86_64
apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                     apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch              httpd-2.4.65-1.amzn2023.0.2.x86_64
httpd-core-2.4.65-1.amzn2023.0.2.x86_64                       httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                      libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch                           mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.65-1.amzn2023.0.2.x86_64                          mod_lua-2.4.65-1.amzn2023.0.2.x86_64

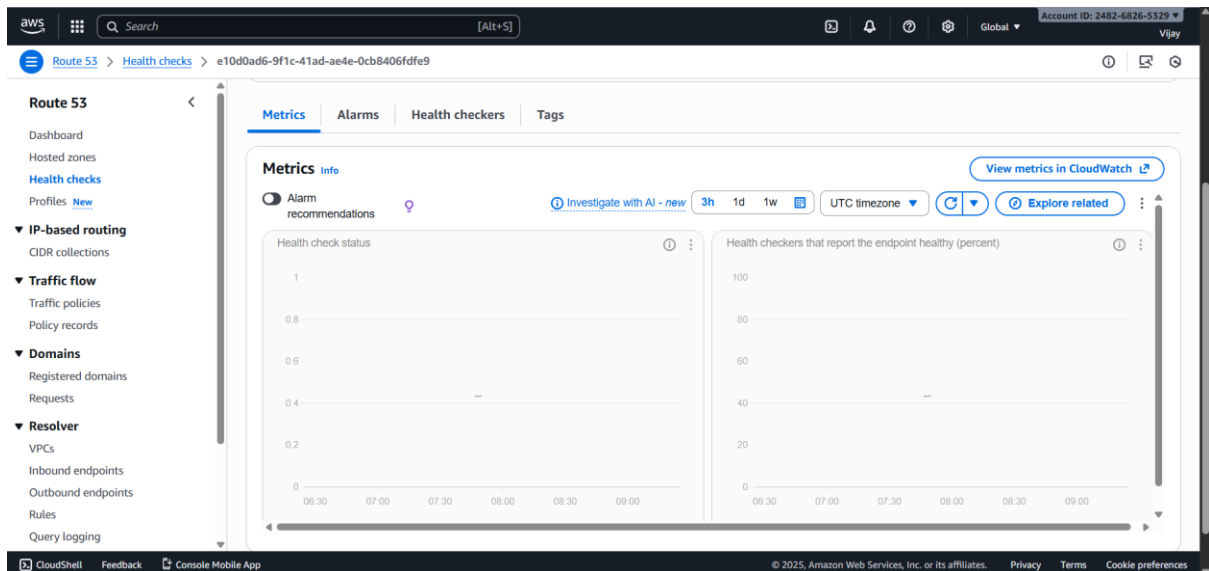
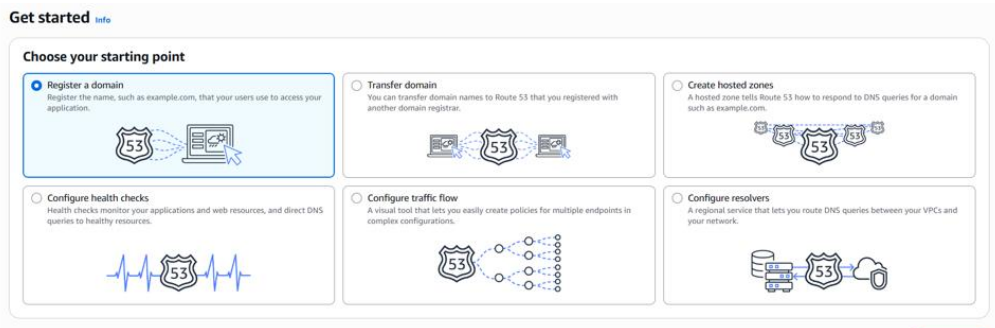
Complete!
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl start httpd
[root@ip-172-31-72-108 ec2-user]# sudo systemctl enable httpd
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-11-14 06:11:07 UTC; 21min ago
     Docs: man:httpd.service(8)
  Main PID: 25542 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1053)
    Memory: 13.3M
       CPU: 1.166s
    CGroup: /system.slice/httpd.service
            └─25542 /usr/sbin/httpd -DFOREGROUND
              └─25543 /usr/sbin/httpd -DFOREGROUND
                └─25544 /usr/sbin/httpd -DFOREGROUND
                  └─25545 /usr/sbin/httpd -DFOREGROUND
                    └─25546 /usr/sbin/httpd -DFOREGROUND

Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal httpd[25542]: Server configured, listening on: port 80
[root@ip-172-31-72-108 ec2-user]#
```



RESULT

Thus, successfully deployed and configured is a foundational web infrastructure: a dynamic EC2/Apache server is operational alongside a scalable, cost-optimized S3 static hosting platform, all centrally managed by AWS Route 53 which provides the domain name, intelligent routing policies, and proactive health checks to ensure high availability and automated failover capabilities while S3 Lifecycle Management optimizes storage costs.

EXP.NO: 04	Create and configure Amazon Elastic File System (EFS) to create scalable and shared file storage for AWS resources and EFS file system, mount it to EC2 instances, and configure access permissions
DATE: 13-11-25	

AIM

To successfully create, configure, and securely mount an Amazon Elastic File System (EFS) file system onto one or more Amazon EC2 (Linux) instances within the same Virtual Private Cloud (VPC), thereby establishing a highly available, scalable, and shared network file storage solution for distributed applications. The aim also includes setting up access control using Security Groups and, optionally, EFS Access Points and IAM policies.

PROCEDURE

1. Create EFS File System: Navigate to the EFS console, select "Create file system," choose your target VPC, and configure performance/storage settings.
2. Define Mount Targets: When creating the EFS, select the appropriate Availability Zones (AZs) to automatically create Mount Targets and attach a new or existing Security Group (e.g., EFS-SG) to these targets.
3. Configure EFS Security Group (EFS-SG): Edit the inbound rules of the EFS-SG to explicitly allow NFS traffic (Port 2049), specifying the Security Group of your EC2 instance(s) as the source.
4. Verify EC2 Security Group (EC2-SG): Ensure the EC2-SG has an outbound rule that permits all traffic (or specifically Port 2049) destined for the EFS-SG to allow the connection initiation.
5. Connect and Install Utility: SSH into your EC2 instance and install the recommended Amazon EFS client utility (amazon-efs-utils) or the generic nfs-common package if using a non-Amazon Linux distribution.

6. Create Mount Point: Create a local directory on the EC2 instance to serve as the mount point (e.g., /mnt/efs_share).
7. Mount the File System: Use the EFS Mount Helper command (found in the EFS console's "Attach" instructions) to execute the mount command: `sudo mount -t efs -o tls fs-xxxxxxx:/mnt/efs_share`.

IMPLEMENTATION SCREENSHOT

The top screenshot displays the AWS Management Console's EC2 Instances page. A table lists instances, with the following data visible:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
23bai10917...	i-092f429fc8cc79424	Running	t3.micro	Initializing	View alarms +	us-east-1f	ec2-44-210-134-91.co...	44.210

The bottom screenshot shows the Amazon EFS File systems page. A table lists file systems, with the following data visible:

Name	File system ID	Encrypte d	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)	File system state	Creation time
23bai10917-elastic-file-system	fs-09eaabfd43cd3ab6e	Encrypte d	0 Bytes	0 Bytes	0 Bytes	0 Bytes	-	Creating	Wed, 12 Nov 2025 15:51:48 GMT

23bai10917-elastic-file-system (fs-09eaabfd43cd3ab6e)

General

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:248268265329:file-system/fs-09eaabfd43cd3ab6e

Automatic backups
Enabled

Encrypted
3b6ce6f1-7392-4310-9fec-f1cef1cd927 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com

Replication overwrite protection
Enabled

Metered size
Total size
6.00 KiB

Metered size

Total size
6.00 KiB

Size in Standard
6.00 KiB (100%)

Size in IA
0 Bytes (0%)

Size in Archive
0 Bytes (0%)

6.00 KiB

Size in Standard
6.00 KiB

Size in Standard Size in IA Size in Archive

Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS ☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-09eaabfd43cd3ab6e:/ efs
```

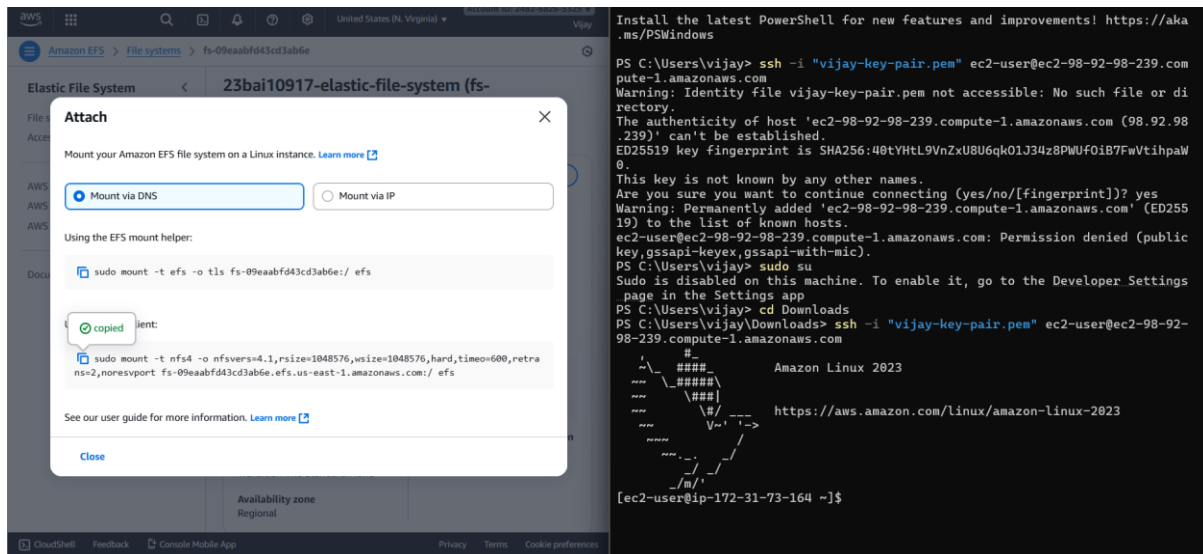
Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsizex=1048576,hard,timeo=600,retrns=2,noresvport fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com:/ efs
```

See our user guide for more information. [Learn more](#)

Close

Metered size
Total size
6.00 KiB



RESULT

The EFS file system will be successfully mounted to the EC2 instance, verified by the presence of the EFS network volume when running the `df -hT` command. This result confirms that the storage is decoupled from the compute instance, allowing files written to the mounted directory to be immediately accessible, scalable, and persistent across all other EC2 instances securely mounted to the same EFS.

EXP.NO: 05	Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience
DATE: 13-11-25	

AIM

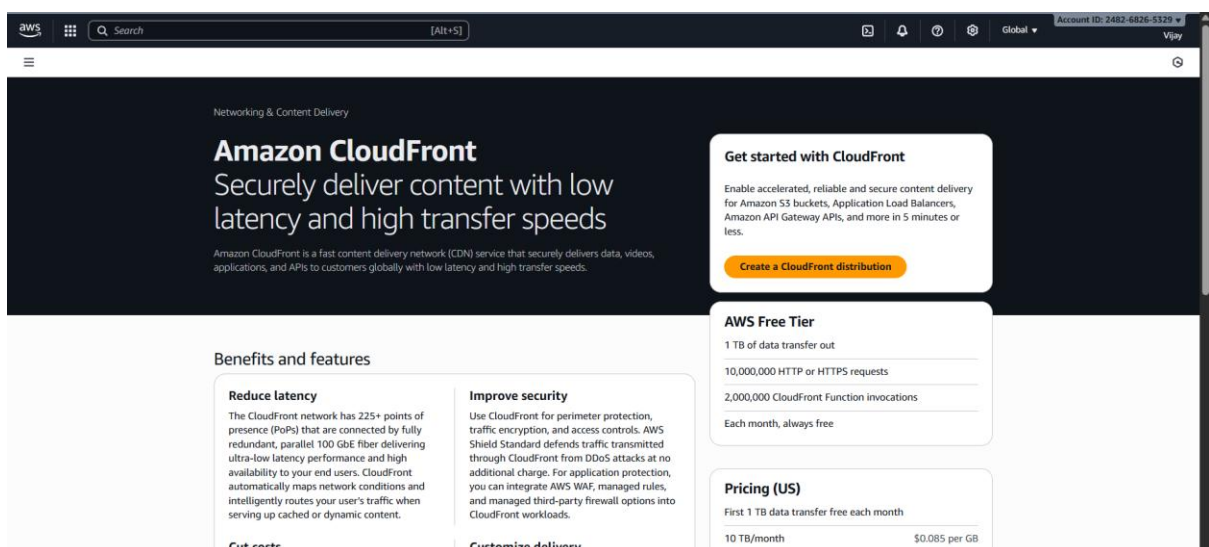
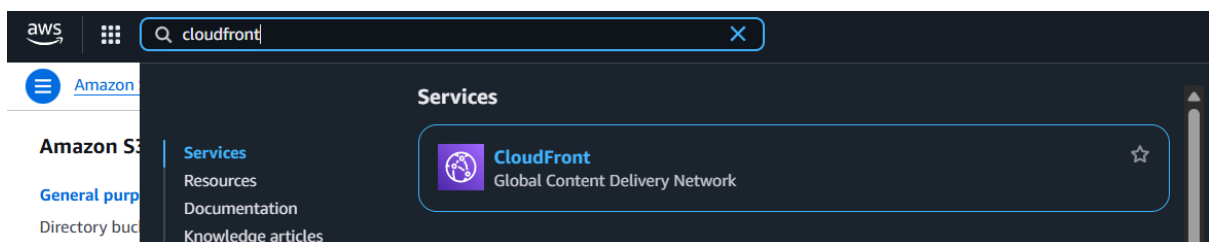
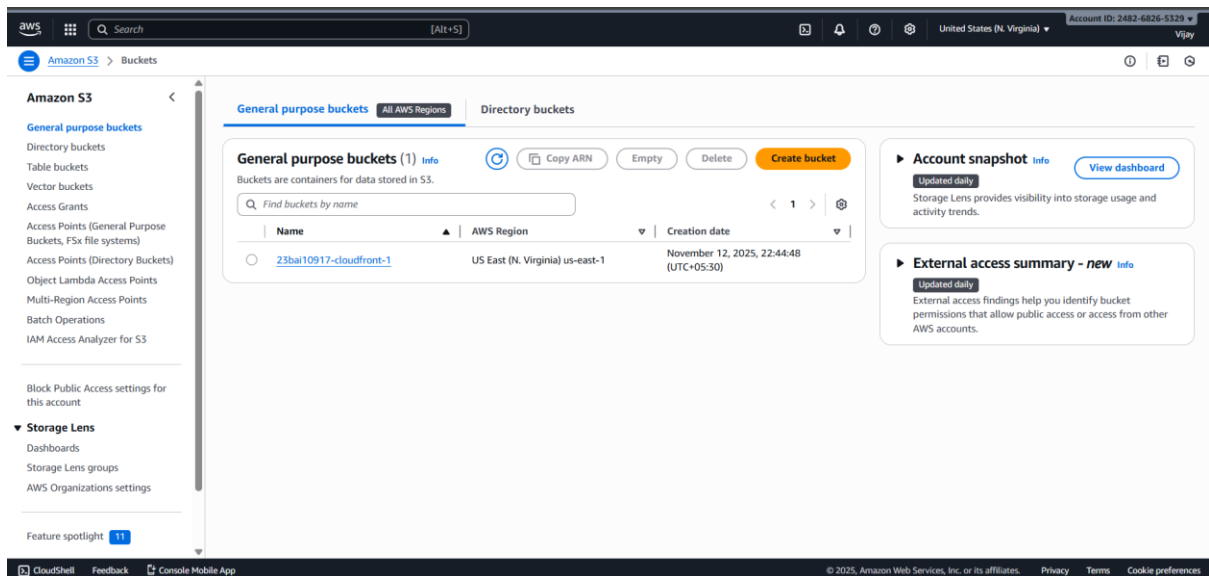
To Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience.

PROCEDURE

1. Select Distribution Type: Navigate to the CloudFront console and choose "Create Distribution," selecting the appropriate delivery method (usually Web for website content).
2. Define Origin: Specify the Origin Domain Name (e.g., your S3 bucket endpoint or Load Balancer DNS for EC2), set a unique Origin ID, and configure an Origin Access Control (OAC) to restrict direct public access to the Origin.
3. Configure Cache Behavior: Define the path patterns (e.g., Default (*)) and set the Viewer Protocol Policy (e.g., Redirect HTTP to HTTPS) and Allowed HTTP Methods (e.g., GET, HEAD).
4. Set Caching Policy: Choose or create a Cache Policy to control how long content stays in the edge locations (Time-To-Live or TTL) and which request headers, cookies, or query strings are forwarded to the Origin.
5. Distribution Settings: Specify the Price Class (geographic regions to serve), the Alternate Domain Names (CNAMEs) for your custom domain, and the Custom SSL Certificate for secure connections.
6. Create and Wait: Review all settings and click "Create Distribution." Wait for the distribution's status to change from InProgress to Deployed (this typically takes 15–30 minutes).
7. Update DNS: Once deployed, update your domain's DNS records by creating a CNAME record that points your custom domain name (e.g., www.example.com) to the new CloudFront Distribution Domain Name.

IMPLEMENTATION SCREENSHOT

Creating S3 bucket.



Successfully created new distribution.

23bai10917-cloudfront-distribution Standard View metrics

Details

Distribution domain name duo4nz3fjpngx.cloudfront.net	ARN arn:aws:cloudfront::248268265329:distribution/E1OH9NKFTL28IE	Last modified Deploying
--	---	----------------------------

General | Security | Origins | Behaviors | Error pages | Invalidations | Tags | Logging

Settings Edit

Name 23bai10917-cloudfront-distribution Description - Price class Use all edge locations (best performance) Supported HTTP versions HTTP/2, HTTP/1.1, HTTP/1.0	Alternate domain names - Add domain	Standard logging Off Cookie logging Off Default root object -
---	---	---

CloudFront > Distributions > EBP8YJN3KMYV > Edit origin

CloudFront

Distributions
Policies
Functions
Static IPs
VPC origins
What's new

Bucket can restrict access to only CloudFront.

☐ Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.

Origin access control
Select an existing origin access control (recommended) or create a new control.

23bai10917-cloudfront-1.s3.us-east-1.amazonaws.com Create new OAC

ⓘ You must allow access to CloudFront using this policy statement. Learn more about [giving CloudFront permission to access the S3 bucket](#). Copy policy

Amazon S3 > Buckets > 23bai10917-cloudfront-1 > Edit bucket policy

Amazon S3

General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens
Dashboards
Storage Lens groups
AI/ML Optimization settings

Bucket ARN
arn:aws:s3::23bai10917-cloudfront-1

Policy

```

1  {
2    "Version": "2008-10-17",
3    "Id": "PolicyForCloudFrontPrivateContent",
4    "Statement": [
5      {
6        "Sid": "AllowCloudFrontServicePrincipal",
7        "Effect": "Allow",
8        "Principal": {
9          "Service": "cloudfront.amazonaws.com"
10       },
11       "Action": "s3:GetObject",
12       "Resource": "arn:aws:s3:::23bai10917-cloudfront-1/*",
13       "Condition": {
14         "StringEquals": {
15           "AWS:SourceArn": "arn:aws:cloudfront::248268265329:distribution/EBP8YJN3KMYV"
16         }
17       }
18     ]
19   }
20 }

```

Edit statement

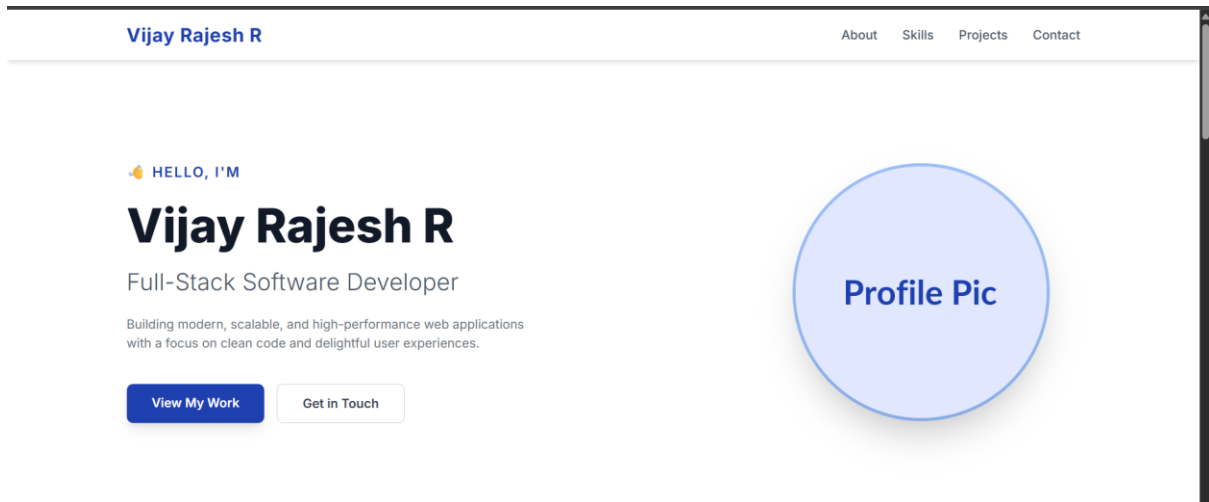
Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

23bai10917-cloudfront-1

Successfully edited bucket policy.



RESULT

Upon updating the DNS and successful propagation, website visitors accessing the custom domain will be automatically routed to the nearest CloudFront Edge Location, resulting in reduced latency, faster loading times, and a highly available content delivery experience. The distribution will effectively cache content, minimizing direct requests to the Origin and improving overall website performance and security.